KYLE MALONEY

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EDUCATION

Iowa State University — Ames, IA *May 2026*
Bachelor of Science in Software Engineering, Minor in Cybersecurity GPA: 3.0
Relevant Coursework: Cloud Computing, Network Security, Cryptography, Operating Systems, Data Structures & Algorithms, Software Design

TECHNICAL SKILLS

Cloud & Infrastructure: AWS (S3), Azure, Google Cloud (GCP), Kubernetes (MicroK8s), Linux server administration, container orchestration
CI/CD & Monitoring: GitHub Actions, Keel, Grafana, Prometheus, Git, Bash
Networking & Security: Cloudflare Tunnel (DNS, proxying, TLS), AdGuard Home, network configuration, Wireshark, Nmap, applied cryptography
Languages & Databases: Python, Bash, SQL, Java, C, C++, JavaScript, MySQL, SQLite, NoSQL

INFRASTRUCTURE EXPERIENCE

Self-Hosted Homelab — Kubernetes Application & Service Platform *Dec 2023 – Present*
- Design, deploy, and maintain a self-managed MicroK8s (Kubernetes) cluster on Linux that hosts production-style services: a media server, network-attached storage (NAS), photo management, a file manager, and public websites including my portfolio (thekyle.net) and blog.
- Built CI/CD pipelines with GitHub Actions and Keel that build the portfolio site and blog on every push and automatically roll out updated pods to the cluster.
- Expose internet-facing services through Cloudflare Tunnel — public DNS, proxying, and TLS with no open inbound ports — and run internal DNS with AdGuard Home for network-wide ad and threat filtering.
- Monitor infrastructure health with Prometheus metrics and Grafana dashboards; manage deployments, persistent storage, and service uptime for users worldwide.
- Handle end-to-end operations: Linux administration, networking, troubleshooting, and recovery from hardware and service failures.

PROJECTS

GridSAFE — AI-Based Anomaly Detection for Power-Grid Networks | *Senior Capstone* *Aug 2025 – May 2026*
- Built a simulated power-grid network to generate realistic traffic and anomaly data, then trained an XGBoost anomaly-detection model that caught 80% of anomalies in the test environment, flagging faults and intrusions on critical infrastructure.
- Collaborated across a multi-disciplinary 5-person team through the full lifecycle: network design, data generation, model training, and validation.

Secure Vault — Zero-Knowledge Encrypted Cloud Storage on AWS *Feb 2026 – May 2026*
- Designed the cloud architecture for a zero-knowledge storage app: a Flask API backed by AWS S3 (boto3) that stores only client-side AES-256-GCM ciphertext — file contents, filenames, and keys stay unreadable even if the provider is breached.
- Implemented envelope encryption with per-file wrapped data keys, allowing master-password rotation in a single atomic server transaction without touching a byte of ciphertext in the cloud.
- Built resumable, chunked transfers resilient to crashes and network failures, using a local state ledger to track sync state and avoid redundant work.

LED-Flux — Distributed Real-Time LED Control System *Dec 2025*
- Deployed a three-process service stack on Raspberry Pi — React frontend, Flask API, and a Python render engine — separated over UDP IPC so the API can restart independently without disturbing the real-time render loop.
- Wrote shared configuration and startup scripts for repeatable deployment, and engineered the engine to degrade gracefully: malformed commands and failing animation layers are logged and dropped rather than crashing the service.

EXPERIENCE

Dining Hall Supervisor — Iowa State University Dining, Ames, IA *Oct 2022 – Feb 2026*
- Supervised a team of student staff during high-volume dining service, coordinating daily operations and resolving service issues in real time while upholding health, safety, and quality standards.
- Created individualized weekly plans for students across multiple workstations to accomplish tasks efficiently and effectively.